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Mini Project Report on

“Smart City: Noise Heat Mapping”

Submitted in partial fulfillment of the requirements for the
Second Semester of the Bachelor of Engineering Degree, towards the completion of the
Mini Project under the **Interdisciplinary Project based learning**,
Department of Basic Sciences.

by

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CERTIFICATE

This is to certify that the File Structures mini project entitled “Smart City: Noise Heat Mapping” has been successfully carried out by Janhavi Dev(1CR25CI021), Jasleen Kaur(1CR25CI022), Rounit Raj (1CR25AD093), S Hruday Simha(1CR25AD094), bonafide students of **CMR Institute of Technology**.

The project is submitted in partial fulfillment of the requirements for the First Semester of the Bachelor of Engineering Degree, towards the completion of the Mini Project under the **Innovation & Design Thinking Laboratory, Department of Basic Sciences**.

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This File Structures mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

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Abstract

This project presents a smart urban intelligence platform that visualizes and analyzes city-wide noise pollution using interactive heatmaps integrated with traffic congestion and temperature data. The primary objective is to support citizens, urban planners, and traffic management authorities in making informed decisions related to transportation, housing, and sustainable city development. The system uses a web-based architecture that combines geospatial visualization, real-time data synchronization, and AI-assisted analytics to identify congestion hotspots, monitor environmental stressors, and evaluate urban livability conditions. Interactive dashboards developed using modern frontend technologies provide dynamic layered maps displaying decibel intensity and mobility patterns across different city regions. Additionally, the platform includes a user-oriented feature that enables house and area selection based on budget, environmental comfort, and preferred noise levels. The proposed solution aims to improve urban planning efficiency, reduce exposure to harmful noise pollution, and enhance public awareness regarding environmental quality. Future enhancements include AI-based predictive analysis, integration with IoT sensor networks, and intelligent route optimization for quieter and less congested travel.

Keywords: Noise Heatmap, Smart City Analytics, Traffic Congestion, Urban Intelligence, Geospatial Visualization, AI-Based Prediction

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Chapter 1: Introduction

1.1 Brief Introduction to SONIC-GRID

SONIC-GRID is a Smart Noise Pollution Monitoring and Visualization System developed to address the growing problem of urban noise pollution. The system provides real-time visualization of noise levels using interactive maps, heatmaps, and citizen reporting features. It helps users identify high-noise zones and promotes public awareness regarding safe sound levels.

The project uses modern web technologies such as React.js, Leaflet.js, JavaScript, HTML, and CSS to create an interactive and user-friendly platform. The system enables users to monitor urban noise activity, report noisy areas, and analyze data visually through dashboards and map-based interfaces.

1.2 Primary Goals for Design of the System

The primary goals of the SONIC-GRID system are:

- To visualize urban noise pollution in real time
- To create an interactive map-based monitoring system
- To provide citizen reporting functionality
- To increase public awareness regarding noise pollution
- To develop a responsive and user-friendly interface
- To provide analytical insights using dashboards and heatmaps

Chapter 2: About the project

2.1 Problem Statement

Urban areas are increasingly affected by noise pollution due to traffic, construction work, industrial activities, and public gatherings. Existing systems do not provide accessible real-time visualization of noise levels, and citizens lack a simple platform to report noisy locations. This creates a need for a smart and interactive monitoring system.

2.2 Objective

The objectives of the project are:

- To monitor and visualize noise pollution data
 - To identify noise hotspots using heatmaps
 - To provide a citizen reporting mechanism
 - To improve public awareness regarding harmful noise levels
 - To develop an interactive analytics dashboard
-

2.3 Solution

SONIC-GRID solves the problem by providing:

- A live interactive map for monitoring noise levels
- Dynamic hotspot markers showing noisy locations
- Heatmap visualization for identifying high-noise zones
- Citizen reporting functionality for community participation
- Dashboard analytics for better understanding of noise trends

The system uses frontend technologies and mapping libraries to create a scalable and responsive web application.

2.4 Advantages

Advantages of SONIC-GRID include:

- Real-time noise visualization
- Improved public awareness
- Interactive and easy-to-use interface
- Community-driven reporting system
- Scalable architecture for future IoT integration
- Responsive dashboard and map controls

Chapter 3: Design

3.1 Description of Approach to Solve the Problem

The project follows a web-based visualization approach using interactive maps and dashboards. The system displays noise data using markers and heatmaps on a digital map. Users can report noisy locations, and the system updates visual indicators dynamically.

The overall workflow includes:

1. Collecting or simulating noise data
 2. Displaying data on the interactive map
 3. Generating heatmap layers for visualization
 4. Allowing users to report noisy zones
 5. Showing analytics and trends on dashboards
-

3.2 Steps to Implement the System

Step 1: Project Setup

- Created project using Vite and React.js
- Configured dependencies and folder structure

Step 2: UI Development

- Designed responsive user interface using HTML and CSS
- Created dashboard and navigation components

Step 3: Map Integration

- Integrated Leaflet.js for interactive maps
- Added map controls and markers

Step 4: Heatmap Visualization

- Implemented heatmap layer for noise intensity visualization

Step 5: Citizen Reporting Module

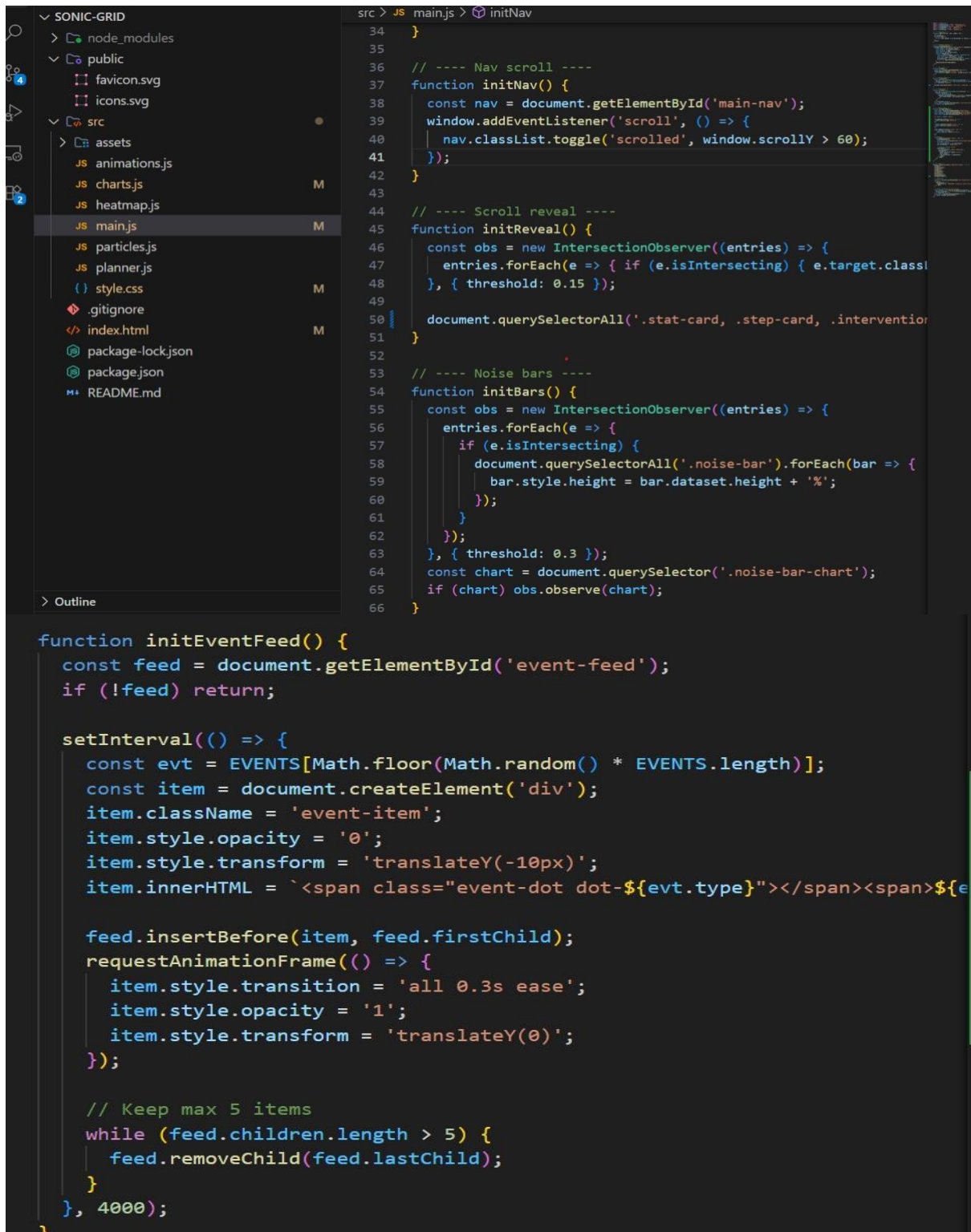
- Added forms for reporting noisy locations
- Displayed reports dynamically on the map

Step 6: Testing and Refinement

- Conducted user testing
- Improved responsiveness and UI interactions

Chapter 4: Code snippets and Descriptions

4.1 Code snippets for Noise Monitoring Map



```
src > JS main.js > initNav
34 }
35
36 // ---- Nav scroll ----
37 function initNav() {
38   const nav = document.getElementById('main-nav');
39   window.addEventListener('scroll', () => {
40     nav.classList.toggle('scrolled', window.scrollY > 60);
41   });
42 }
43
44 // ---- Scroll reveal ----
45 function initReveal() {
46   const obs = new IntersectionObserver((entries) => {
47     entries.forEach(e => { if (e.isIntersecting) { e.target.class
48   }, { threshold: 0.15 });
49
50   document.querySelectorAll('.stat-card, .step-card, .intervention
51 }
52
53 // ---- Noise bars ----
54 function initBars() {
55   const obs = new IntersectionObserver((entries) => {
56     entries.forEach(e => {
57       if (e.isIntersecting) {
58         document.querySelectorAll('.noise-bar').forEach(bar => {
59           bar.style.height = bar.dataset.height + '%';
60         });
61       }
62     });
63   }, { threshold: 0.3 });
64   const chart = document.querySelector('.noise-bar-chart');
65   if (chart) obs.observe(chart);
66 }

function initEventFeed() {
  const feed = document.getElementById('event-feed');
  if (!feed) return;

  setInterval(() => {
    const evt = EVENTS[Math.floor(Math.random() * EVENTS.length)];
    const item = document.createElement('div');
    item.className = 'event-item';
    item.style.opacity = '0';
    item.style.transform = 'translateY(-10px)';
    item.innerHTML = `<span class="event-dot dot-${evt.type}"></span><span>${e

    feed.insertBefore(item, feed.firstChild);
    requestAnimationFrame(() => {
      item.style.transition = 'all 0.3s ease';
      item.style.opacity = '1';
      item.style.transform = 'translateY(0)';
    });

    // Keep max 5 items
    while (feed.children.length > 5) {
      feed.removeChild(feed.lastChild);
    }
  }, 4000);
}
```

4.2 Generalized Code Description

The above code demonstrates the implementation of an interactive noise monitoring map using React.js and Leaflet.js.

Description:

- MapContainer initializes the interactive map.
- TileLayer loads OpenStreetMap tiles.
- Noise locations are stored in an array.
- Marker displays hotspot locations on the map.
- Popup shows information about the noise level.

This code forms the foundation of the live noise visualization system.

Chapter 5: Result and Analysis

5.1 Result

The SONIC-GRID prototype successfully demonstrated:

- Real-time noise hotspot visualization
- Interactive map functionality
- Heatmap integration
- Citizen reporting capability
- Responsive analytics dashboard

Users found the system informative, user-friendly, and effective in identifying noisy urban areas.

5.2 Analysis

System Performance Analysis

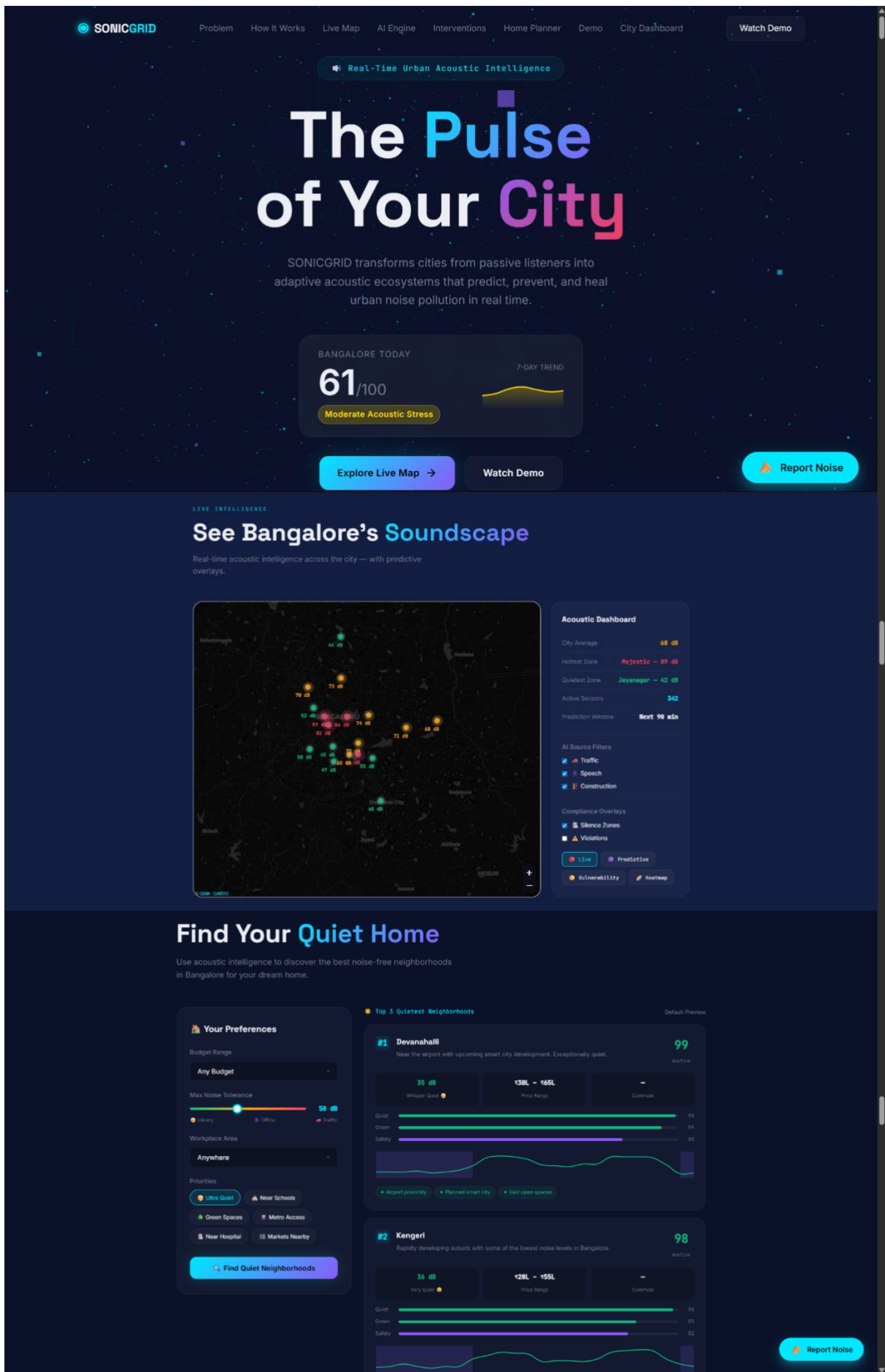
The system performed efficiently during testing and demonstrated smooth rendering of maps, animations, and UI components.

Observations:

- Interactive maps loaded correctly without significant delay
- Heatmap visualization responded dynamically
- Event feed updates worked continuously at fixed intervals
- Scroll animations improved user interaction
- Dashboard elements remained responsive across different screen sizes

The use of lightweight frontend technologies helped maintain smooth performance and fast loading times.

5.3 Screenshots:



Chapter 6: Conclusion

SONIC-GRID successfully addresses the issue of urban noise pollution through interactive visualization and citizen participation. The project demonstrates how modern web technologies can be used to build real-time monitoring systems for public awareness and urban management.

The system provides an efficient platform for visualizing noise hotspots, reporting noisy areas, and analysing trends through dashboards and heatmaps. Future improvements may include real IoT sensor integration, AI-based noise prediction, and enhanced scalability for smart city applications.

Chapter 7: Reference

1. [React.js Documentation](#)
 2. [Leaflet.js Official Documentation](#)
 3. [OpenStreetMap API Documentation](#)
 4. [JavaScript MDN Web Docs](#)
 5. [HTML & CSS Documentation](#)
 6. [GitHub Documentation](#)
 7. [Urban Noise Pollution Research Articles](#)
- [sVite Official Documentation](#)